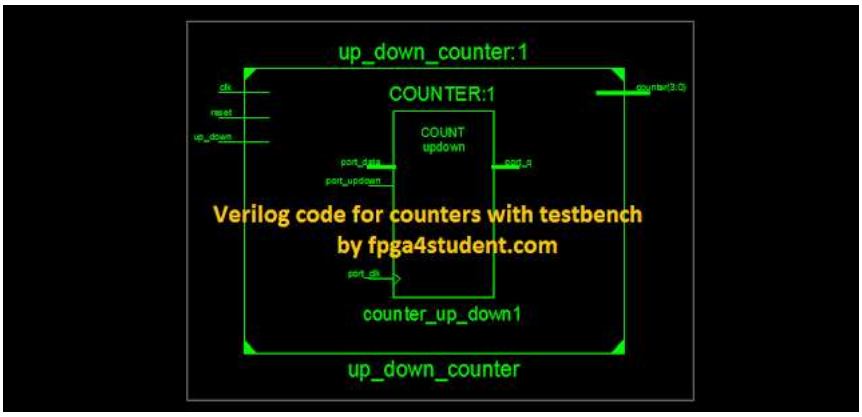
Verilog code for counter with testbench

In this project, Verilog code for counters with testbench will be presented including up counter, down counter, up-down counter, and random counter.



Verilog code for up counter:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// Verilog code for up counter
module up_counter(input clk, reset, output[3:0] counter
    );
reg [3:0] counter_up;

// up counter
always @(posedge clk or posedge reset)
begin
if(reset)
 counter_up <= 4'd0;
else
 counter_up <= counter_up + 4'd1;
end
assign counter = counter_up;
endmodule
```

Verilog testbench code for up counter:

```
// FPGA projects using Verilog/ VHDL
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// Verilog code for up counter with testbench
// Testbench Verilog code for up counter
module upcounter_testbench();
reg clk, reset;
wire [3:0] counter;

up_counter dut(clk, reset, counter);
initial begin
clk=0;
forever #5 clk=~clk;
end
initial begin
reset=1;
#20;
reset=0;
end
endmodule
```
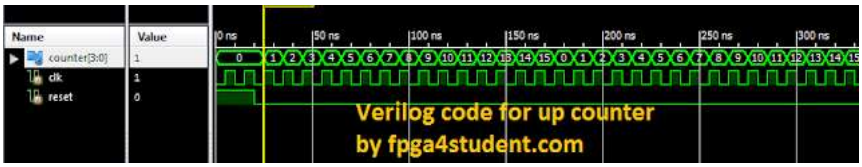

Simulation waveform for up counter:



Verilog code for down counter:

```
// FPGA projects using Verilog/ VHDL
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// Verilog code for down counter
module down_counter(input clk, reset, output [3:0] counter
    );
reg [3:0] counter_down;

// down counter
always @(posedge clk or posedge reset)
begin
if(reset)
 counter_down <= 4'hf;
else
 counter_down <= counter_down - 4'd1;
end
assign counter = counter_down;
endmodule
```

Verilog testbench code for down counter:

```
// FPGA projects using Verilog/ VHDL
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// Verilog code for down counter with testbench
// Testbench Verilog code for down counter
module downcounter_testbench();
reg clk, reset;
wire [3:0] counter;

down_counter dut(clk, reset, counter);
initial begin
clk=0;
forever #5 clk=~clk;
end
initial begin
reset=1;
#20;
reset=0;
end
endmodule
```
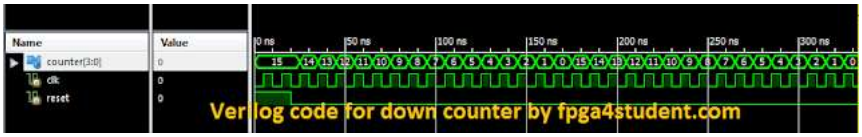

Simulation waveform for down counter:



Verilog code for up-down counter:

```
// FPGA projects using Verilog/ VHDL
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// Verilog code for up-down counter
module up_down_counter(input clk, reset,up_down, output[3:0]  counter
    );
reg [3:0] counter_up_down;
```

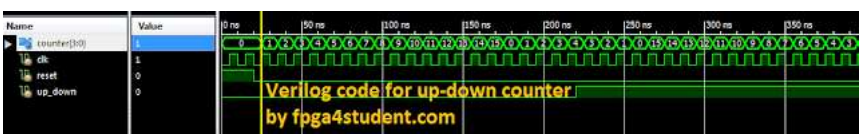
```
// down counter
always @(posedge clk or posedge reset)
begin
if(reset)
 counter_up_down <= 4'h0;
else if(~up_down)
 counter_up_down <= counter_up_down + 4'd1;
else
 counter_up_down <= counter_up_down - 4'd1;
end
assign counter = counter_up_down;
endmodule
```

Verilog testbench code for up-down counter:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// Verilog code for up-down counter with testbench
// Testbench Verilog code for up-down counter
module updowncounter_testbench();
reg clk, reset,up_down;
wire [3:0] counter;

up_down_counter dut(clk, reset,up_down, counter);
initial begin
clk=0;
forever #5 clk=~clk;
end
initial begin
reset=1;
up_down=0;
#20;
reset=0;
#200;
up_down=1;
end
endmodule
```

Simulation waveform for up-down counter:



Verilog code for random counter using LFSR:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// Verilog code for random counter using linear shift feedback register
 module random_counter_lfsr(input clk, rst_n,
                 input[4:0] initialized_value,
                 output[4:0] counter_random);
 wire [4:0] counter_lfsr;
 wire d_xor;
  xor xor_u(d_xor,counter_lfsr[1],counter_lfsr[4]);
 D_FF u0(.q(counter_lfsr[0]), .d(counter_lfsr[4]), .rst_n(rst_n), .clk(clk),.init_
 D_FF u1(.q(counter_lfsr[1]), .d(counter_lfsr[0]), .rst_n(rst_n), .clk(clk),.init_
 D_FF u2(.q(counter_lfsr[2]), .d(d_xor), .rst_n(rst_n), .clk(clk),.init_value(init
 D_FF u3(.q(counter_lfsr[3]), .d(counter_lfsr[2]), .rst_n(rst_n), .clk(clk),.init_
 D_FF u4(.q(counter_lfsr[4]), .d(counter_lfsr[3]), .rst_n(rst_n), .clk(clk),.init_
 assign counter_random = counter_lfsr;
 endmodule
 // FPGA projects using Verilog/ VHDL
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
```

```
// Verilog code for random counter using linear shift feedback register
// Verilog code for D_FF using in random counter
 module D_FF (q, d, rst_n, clk,init_value);
 output q;
 input d, rst_n, clk,init_value;
 reg q;
 always @(posedge clk or negedge rst_n)
 if (~rst_n)
 q <= init_value;
 else
 q <= d;
 endmodule
```

Verilog testbench code for random counter using LFSR:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// Verilog code for random counter with testbench
// Testbench Verilog code for random counter
module randomcounter_testbench();
reg clk, reset;
reg [4:0] initialized_value;
wire [4:0] counter_random;

random_counter_lfsr dut( clk, reset,
                 initialized_value,
                 counter_random);
initial begin
clk=0;
forever #5 clk=~clk;
end
initial begin
reset=0;
initialized_value=5'b11111;
#20;
reset=1;
end
endmodule
```

Simulation waveform for random counter:

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